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Research article

Optimizing inventory level and technology investment under a carbon tax, cap-and-trade and strict carbon limit regulations

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ABSTRACT

In this study, three models are developed to optimize the inventory level and technology investment under a carbon tax, cap-and-trade, and strict carbon limit regulations. The study considers warehouse operations and transportation logistic systems as the sources of emissions. Several examples are presented, followed by sensitivity analysis for different scenarios. The result shows that under the carbon regulations, profit and carbon emission reduction is proportional to the green technology investment, while demand increase with the green technology promotion. Further analysis has shown that for the carbon tax system policy, the carbon price has influenced the total profit significantly. The decreasing emission level results in a higher profit, while the carbon cap and the carbon limit do not affect the total profit significantly.

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1. Introduction

Since the signing of the Paris Agreement in 2016, many countries have made efforts to reduce the harmful effects of carbon emissions which result in climate change and global warming. Governments and world organizations are accelerating and intensifying the many investment programs and projects needed to sustain low-carbon development. The World Bank (2020) reported more than 61 carbon pricing mechanisms including a carbon tax and cap-and-trade systems around the world. A carbon tax system is set by a government to impose greenhouse gas (GHG) emission limits to the enterprises; it is based on the amount of carbon dioxide (CO_2) emitted from their operations. In a cap-and-trade system, the government sets limits on the CO_2 emissions of companies and regulates emission allowances trading between companies. Moreover, a carbon limit/cap system strictly set an upper limit of emissions that companies may not exceed.

Besides regulations, governments also encourage industries to implement green industry practices. Therefore, under various carbon regulations, industries seek efficient methods to reduce emissions by investing in more environmentally friendly technologies and improved operations planning (Bouchery et al., 2012; Benjaafar et al., 2013). Besides incorporating emission costs in their operational decisions, companies also invest in greener technologies to reduce emissions and comply with the limits. For example, The Home Depot (2018) reported emissions reduction through renewable and alternative energy source usage, while Panasonic Group (2020) consolidates product deliveries, investing in low-emission vehicles, and promoting biofuel consumption. These investments increase the company's expenses. However, due to customers' awareness and their increasing willingness to buy greener products, it will compensate the extra costs of investing in green technology (Zanoni et al., 2014; Lou et al., 2015; Xia et al., 2018). In order to increase customer awareness, a good promotion program is necessary (Xia et al., 2018). The above studies motivate us to investigate the optimal investment and inventory levels when the industry uses promotions to increase customer demand. Since all companies are bound by government regulations regarding carbon policies, the objective of our research is to maximize profit, implement green technology, reduce carbon emissions and fulfill environmental responsibility.

Inventory (stock or lot sizing) management is a challenging logistics operation because the decisions greatly affect a company's financial and service performance. Lately, researchers and practitioners have also explored how inventory management affects a company's environmental performance, leading to investigations on sustainable or low-carbon inventory management. In-

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ventory management most directly influences transportation and storage activities. Both are closely related to resources and energy (e.g., fuel and electricity) consumption which directly or indirectly increases the rate of carbon emissions. Bonney and Jaber (2011) identified vehicle emissions as a social cost that becomes an additional cost in an economic order quantity (EOQ) model. Chen et al. (2013) considered emission costs from the ordering, storage, and purchase activities. Based on a life cycle assessment (LCA) approach, Battini et al. (2014) incorporated carbon emissions which come from fuel generation, soil, electricity, and other energy consumptions from material purchasing, storage, until waste disposal.

Several studies developed sustainable inventory decision-making models based on carbon tax mechanisms. Other sustainable inventory management studies investigated and compared the application of carbon tax, cap-and-trade, and strict carbon cap regulations. This study considers the preceding regulations as well as other aspects of carbon emissions reduction strategy, such as the implementation of green technology and promoting green practice. Toptal et al. (2014) examined the optimum annual green technology investment under different carbon mechanisms. However, they assumed a constant demand pattern. In this study, we investigate different scenarios on the effect of green technology investment on customer demand. Moreover, our model also considers the effect of promoting the company's green performance. Due to the increased customer demand, the investment costs in green technology and promotion can be compensated, hence maintaining the company's profit (Zanoni et al., 2014; Xia et al., 2018).

In summary, since inventories are associated with carbon emissions from production, distribution, and post-production processes, it is necessary to consider sustainable inventory management. In order to reduce its emissions level and achieve sustainability, the related functions need to invest in green technology. Hence, the main contribution of this study is as follows:

i This study proposes a method to derive the optimal order quantity and green technology investment when firms take advantage of the opportunities to promote their environmental performance. The optimum decisions are necessary for firms to maintain their profit.
ii This study provides different decision models that enable retailers to respond to different carbon regulations, i.e. carbon tax, cap-and-trade, and limited carbon cap from the government.
iii This study identifies key factors that affect the total profit in low carbon EOQ models under different carbon regulations. The effect of efficiency and emission factors from the green technology, changes in costs, and carbon regulations' specification are investigated.

Section 2 reviews the relevant literature focusing on the sustainable inventory under carbon emission regulations, green technology investment, and demand function with the effect of green investment and promotion. Section 3 presents the research method by outlining the models. Section 4 presents the results based on several numerical examples and sensitivity analysis. Section 5 provides a discussion based on the findings. Finally, the conclusion and possible future studies are presented in Section 6.

2. Literature review

This section presents the relevant literature on the sustainable inventory model as the basis of this study.

2.1. Sustainable inventory under carbon emission regulations

Responding to various applications of carbon emission regulations, research to reduce carbon emissions in logistics and supply chain inventory systems have been popular since the last decade. Many studies have identified different sources of emissions in the inventory system. Lee (2011) and Wangsa (2017) investigated the direct and indirect carbon footprint of supply chains that relate to transportation activities, while Wahab et al. (2011) emphasized emissions from transportation in an international supply chain. Other studies considered the emissions from transportation and warehouse inventory (Bonney and Jaber, 2011; Hovelaque and Bironneau, 2014; Kazemi et al., 2018; Daryanto et al., 2019; Wee and Daryanto, 2020; Shi et al., 2020; Taleizadeh et al., 2020). For example, a cold supply chain requires temperature-controlled vehicles and warehouses which emits more emissions (Hariga et al., 2017). Therefore, our study considers emissions from transportation and storage due to their large share of emissions.

Researchers have considered different regulations on carbon emissions, i.e. carbon tax, cap-and-trade, and limited carbon emission, on inventory system practices. Benjaafar et al. (2013) identified a need for operational adjustment to respond to different carbon emission regulations and to take advantage of them. Arslan and Turkay (2013) also discussed the social aspect such as work hours besides of different carbon emission regulations. Chen et al. (2013) found a similar emission level when a company operates in a country with a carbon tax or strict emission cap regulations. By introducing a green technology investment, Toptal et al. (2014) showed a reduction in the total carbon emissions level due to the investment under a carbon cap-and-trade and carbon tax policies. Huang et al. (2020) also incorporated the effect of green technology adoption and showed that a carbon tax policy leads to a more efficient investment compared to the other two carbon emission regulations. There are limited industrial research on inventory management under different carbon emission regulations. Hence, our research investigates the simultaneous effect of green technology investment and promotion programs in three different carbon emission regulations to provide governments and industries with insights into their decision making.

2.2. Sustainable inventory with green technology investment

Several studies have examined the possibility of reducing carbon emissions through investment in environmentally friendly technology. Yang and Lin (2020) indicated the importance of green innovation performance for a sustainable supply chain. For example, the supply chain may invest in renewable energy and energy-efficient machines or vehicles. Mishra et al. (2020a) suggested energy-efficient light bulbs to reduce the emissions level in a warehouse operation. Carbon capture technology is another alternative for emission reduction (Shavalieva et al., 2021).

Datta (2017) and Lin (2018) investigated the effect of green technology investment under the carbon tax system focusing on optimizing the investment in transportation activities through renewable energy and equipment acquisition, while Lou et al. (2015) and Xu et al. (2017) worked under the cap-and-trade system. Later, Datta et al. (2019) combined carbon tax and cap-and-trade into one system. Toptal et al. (2014) investigated the optimum annual green technology investment under three different carbon pricing mechanisms. Bhattacharyya and Sana (2019) optimized the green technology of an eco-friendly manufacturing system. Lu et al. (2020) investigated the opportunity of joint green technology investment in a supply chain system. Recently, Mashud et al. (2020b) and Mishra et al. (2020b) incorporated green technology and preservation technology investments to sustain the retailer's profit. All the studies identified the benefit of green technology investment in reducing emissions levels under different carbon emission regulations. The preceding researches have inspired our study.

Table 1
Comparison with the previous related works.

Author(s)	Demand	Carbon pricing policies			Decision Variables
		Carbon tax	Cap and trade	Limited carbon emission	
Benjaafar et al. (2013)	Constant	Yes	Yes	Yes	Order size
Toptal et al. (2014)	Constant	Yes	Yes	Yes	Order size, green technology investment
Zanoni et al. (2014)	Depend on the negative impact of price and Positive impact of environmental performance measure	No	No	No	Order size, green technology investment, selling price
Lou et al. (2015)	Depend on the negative impact of Price and Positive impact of emission reduction level	No	Yes	No	Level of emission reduction, selling Price
Jiang and Chen (2016)	Depend on the negative impact of price and Emission level	No	No	No	Order size, green technology investment, selling price
Taleizadeh et al. (2018)	Depend on the negative impact of price and Positive impact of emission reduction level	No	No	No	Level of emission reduction, selling price
Xia et al. (2018)	Depend on the positive impact of emission reduction level and Promotion	No	Yes	No	Level of emission reduction, level of promotion
Aliabadi et al. (2019)	Depend on the positive impact of credit facility and Negative impact of price and emission level	Yes	No	No	Credit period, inventory cycle
Tao and Xu (2019)	Depend on the negative impact of low carbon awareness level	Yes	Yes	No	Order size
This paper	Depend on the positive impact of green technology investment and its promotion	Yes	Yes	Yes	Order size, green technology investment

2.3. Sustainable inventory with green investment and promotion-sensitive demand

Certain research reported the benefits of the green industry, particularly green logistics, in terms of increasing customer willingness to buy. Huang and Rust (2011) recommended businesses to acquire more green technology as consumers have negative preferences on products with resource-intensive consumption. Tao and Xu (2019) affirmed that customers' awareness of carbon emission levels will motivate them to buy from green technology-focused companies. In addition, demand patterns depend on factors such as product type, market sector, and the quality reputation of a country (Nouira et al., 2016). When considering green technology investments, firms can determine the selling price to cover their expenses. Zanoni et al. (2014), Lou et al. (2015), Zheng and Yang (2015), Jiang and Chen (2016), Taleizadeh et al. (2018), and Aliabadi et al. (2019) studied the optimum price and green technology investment with variable customer demand, while Xia et al. (2018) incorporated the effect of a promotion strategy.

For many years, the positive effect of promotion on customer demand has been proved by many researchers (e.g., Shah et al., 2013; Palanivel and Uthayakumar, 2015; Geetha and Udayakumar, 2016; Mashud et al., 2020a). In this research, it is worthy of considering promotion because the sustainability performance promotion attracting more customers and hence, has a positive impact on demand. However, the model by Xia et al. (2018) only considered cap-and-trade regulation. Our study extends the model for common carbon pricing regulations i.e. carbon tax, cap-and-trade, and limited carbon emission. The study optimizes the green technology investment that maximizes the total profit; the demand is assumed to be dependant on the investment and the promotion level. Table 1 summarizes these works considering customer demand function and carbon pricing mechanism.

3. Methods

This study investigates the EOQ model to derive the optimal order quantity and green technology investment. The list of notations for all variables and parameters are as follows:

Notations	Descriptions
i	Index of cases ($i = 1, 2, 3$)
$R(G)$	Carbon reduction function
p	Selling price per unit
D_0	Constant market demand
D_1	Constant coefficient of $R(G)$
a	Efficiency factor of green technology
b	Emission factor of green technology
D	Demand rate
m	Sensitivity of demand toward promotion, where, $0 < m < 1$
v	Promotion level
d	Delivery distance
E	Total emission
E_T	Carbon emission from distribution
E_h	Carbon emission for storing a unit product
C_0	Purchase price per unit
C_1	Carbon tax for unit carbon emissions
C_2	Carbon trading price
C_T	Transport cost per unit
C_h	Unit holding cost
U	Carbon emissions cap/limit in the cap-and-trade policy
W	Carbon emissions limit in the limited emissions policy
T	Replenishment cycle
	Decision variables
Q	Maximum stock per cycle
G	Green investment cost

The EOQ models are formulated under the following assumptions:

a) EOQ models with infinite replenishment in which shortages are not allowed are considered. In this system, all materials are received at the beginning of the inventory cycle in one delivery which is common in the buyer's inventory system as in Bonney and Jaber (2011), Chen et al. (2013), Wee and Daryanto (2020), Hasan et al. (2020), etc.
b) The carbon emission model in Hovelaque and Bironneau (2014) is a very relevant issue for inventory management because there are many events from that carbon are produced. In this arrangement, we addressed some of the reasons like emitting carbons from the warehouse and from the transport system by which inventories are shipped from the supplier to the retailer warehouse. The vehicle can deliver the whole order size, hence the emission only depends on delivery distance and emission rate per unit distance inspired by Bonney and Jaber (2011).

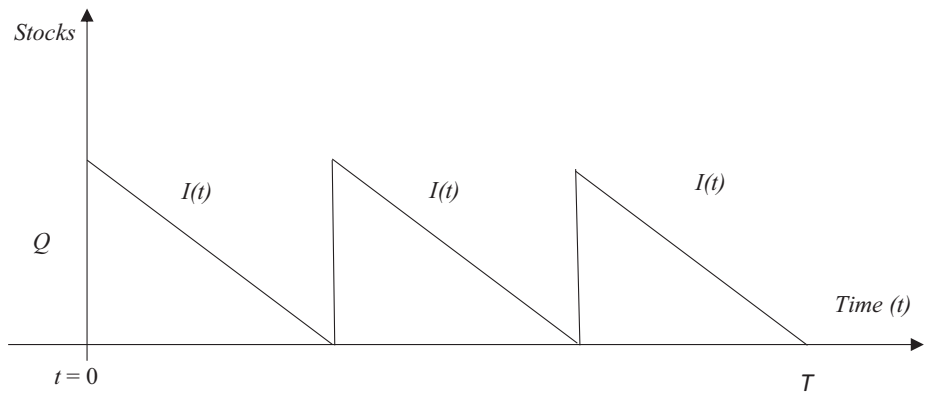

Fig. 1. Graphical representation of the inventory system versus time.

c) Green technology and its appropriate use can make a good result for carbon reduction. The reduction is presented by a function $R(G)$ as follows:
$R(G) = aG - bG^2$, where $G < \frac{a}{b}$ as in Huang and Rust (2011) and Toptal et al. (2014).
The green technology investment G results in an aG reduction of carbon emissions, but adds bG^2 carbon emissions and consumes energy from its operation.

d) Demand is dependent on the green technology investment G as in Zanoni et al. (2014) that can make a positive impact. Also, the promotion level increases customer demand for environmental convenience as in Xia et al. (2018).
This study considers both green technology and its promotion investment. The demand rate is$D(G, v) = D_0 + D_1R(G) + mv$, where $D_0, D_1 > 0$; this means that the carbon emissions reduction and the promotion level v have a positive impact on demand.

e) With all the above concerns interpreted in single-echelon EOQ models with the demand function $D(G, v)$, we consider green technology investment to cut the carbon emissions in the following cases similar to Toptal et al. (2014):
   (i) Carbon taxation policy with investment for green technology.
   (ii) A cap-and-trade policy with investment for green technology.
   (iii) Limited carbon emission policy with investment for green technology.

The following part presents the model formulation. Here, the supplier shifts the products from the production house to the retailer warehouse according to its order. The supplier vehicle covers the distance d and charges the transportation cost to the retailer.

In Fig. 1, the stocks in the retailer's warehouse are at a maximum at $t = 0$ due to the lot size Q. During the time $[0, T]$ the inventory level depletes due to the effect of demand. The stock is zero at T and will be replenished in the next cycle. When a company demonstrates a commitment to sustainability and promotes its products more than its competitors, it impacts sales. Therefore, the inventory system of a retailer's warehouse described by the resulting differential equation and considering demand $D(G, v) = D_0 + D_1R(G) + mv$ is

$$\frac{dI}{dt} = -D(G, v); 0 \le t \le T \tag{1}$$

with $I(t) = Q$ at $t = 0$, and $I(t) = 0$ at $t = T$.

Therefore, by solving Eq. (1), we have the inventory level $I(t)$ and the maximum stocks Q as follows:

$$I(t) = -Dt + DT \tag{2}$$

$$Q = DT \tag{3}$$

3.1. Carbon emissions of the system

Carbon dioxide is emitted from the transportation process, storing process and the use of technologies. The distance traveled by a vehicle is an important factor. In this study, the total carbon emissions for shipping is measured by $\frac{D}{Q}E_Td$, where E_T is the carbon emissions for delivering the products per unit distance during the transportation process, and d is the delivery distance.

Carbon emission from holding management is also considered because the products are stored in the warehouse for a certain period. If the amount of carbon produced for holding one unit of inventory is E_h, then the total emitted carbon for the holding $\int_0^T I(t)dt$ quantity of inventories is:

$$E_h \int_0^T I(t)dt \tag{4}$$

When the corresponding authorities apply carbon pricing regulations and promote carbon emissions reduction, the buyer can invest in green technologies such as solar cell power generation, energy-saving lights, use of less carbon producing vehicles, etc. The function of emissions reduction is

$$R(G) = aG - bG^2, \ G < \frac{a}{b} \tag{5}$$

Therefore, the total carbon emissions from the mentioned process of inventory storage, transportation and green technology investment are

$$E(Q, G) = \frac{D}{Q}E_Td + E_h\int_0^T I(t)dt - R(G) \ = \frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G) \tag{6}$$

3.2. Inventory cost

The inventory model considers ordering cost, purchase cost, transportation cost, holding cost and cost for green technology investment. The order processing cost is referred to as the ordering cost (OC), and the purchase cost includes the cost of transportation. Besides the purchase price, the supplier charges an additional transportation cost per unit product from the retailer. If the original purchase price of each product is C_0 and the cost for transportation of each product from supplier to the retailer is C_T, then the final purchase cost per item is $C_p = C_0 + C_T$ for the retailer. So, the purchase cost incurred to the retailer for Q products is:

$$PC = (C_0 + C_T)Q \tag{7}$$

$$PC = C_pQ \tag{8}$$

The products stay for a certain period in the warehouse leading to a cost for managing the products during the holding period, which is referred to as the holding cost. Thus, the total holding cost is

$$HC = C_h\int_0^T I(t)dt = C_h\frac{DT^2}{2} \tag{9}$$

Finally, to reduce the emissions, this study considers the investment in green technologies G.

3.3. Sales revenue

Sales revenue depends on the item price and quantity of products sold. In this study, we assume all the products are sold within the period $[0, T]$ with a unit selling price p. Hence, the revenue earned from selling the product in time $[0, T]$ is:

$$SR = pDT \tag{10}$$

3.4. Profit function

This section presents the profit functions under different carbon pricing regulations. It also presents some theoretical derivation.

3.4.1. Profit function under carbon tax regulation

The carbon tax is charged proportionally with the total carbon emissions. Hence, from Eq. (6), the total carbon emissions cost under carbon tax policy is

$$TC_1 = C_1[E(Q,\ G)] = C_1\left[\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right] \tag{11}$$

The profit function considering mentioned inventory investments and green technology cost under this case is

$$X_1 = SR - HC - PC - TC_1 - G - OC$$

$$= pDT - C_h\frac{DT^2}{2} - C_pDT - C_1\left[\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right] - G - OC \tag{12}$$

Then the total profit per year is

$$TP_1(Q,G) = \frac{X_1}{T} \Rightarrow TP_1(Q,G)$$
$$= \frac{1}{T}\left[pDT - C_h\left(\frac{DT^2}{2}\right) - C_pDT - C_1\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right) - G - OC\right] \tag{13}$$

After substituting the value of D in Eq. (13) and simplify, we have:

$$TP_1 = \frac{1}{T}\left[\begin{array}{c} pTD_0 - \frac{D_0C_hT^2}{2} - \frac{C_hT^2mv}{2} - C_pTD_0 - C_pTmv + \frac{T^2(E_hC_1D_0+E_hC_1mv)}{2} \\ + pTmv + G\left(pTD_1a - \frac{D_1aC_hT^2}{2} - C_pTD_1a + \frac{aE_hC_1D_1T^2}{2} - C_1a - 1\right) \\ +G^2\left(\frac{C_hT^2}{2}D_1b + C_pTD_1b + C_1b - pTD_1b - \frac{bE_hC_1D_1T^2}{2}\right) \\ -\frac{1}{Q}(C_1E_TdD_0 + C_1E_Tdmv) - \frac{G}{Q}(D_1C_1E_Tda) + \frac{G^2}{Q}D_1C_1E_Tdb \end{array}\right]$$

$$\Rightarrow TP_1 = \frac{1}{T}\left[G\alpha_1 + G^2\alpha_2 - \frac{G}{Q}\alpha_3 + \frac{G^2}{Q}\alpha_4 - \frac{1}{Q}\alpha_5 + \alpha_6\right] \tag{14}$$

where,

$$\alpha_1 = pTD_1a - \frac{D_1aC_hT^2}{2} - C_pTD_1a + \frac{aE_hC_1D_1T^2}{2} - C_1a - 1$$

$$\alpha_2 = \frac{C_hT^2D_1b}{2} + C_pTD_1b + C_1b - pTD_1b - \frac{bE_hC_1D_1T^2}{2}$$

$$\alpha_3 = D_1C_1E_Tda$$
$$\alpha_4 = D_1C_1E_Tdb$$
$$\alpha_5 = C_1E_TdD_0 + C_1E_Tdmv$$

$$\alpha_6 = pTD_0 - \frac{D_0C_hT^2}{2} - \frac{C_hT^2mv}{2} - C_pTD_0 - C_pTmv + \frac{T^2(E_hC_1D_0 + E_hC_1mv)}{2} + pTmv$$

Lemma 1. Under the carbon tax system, the optimal $TP_1(Q, G)$ value is concave. The optimal lot size and green technology cost are:

$$Q^* = \frac{\alpha_2\alpha_3^2 + 4\alpha_2\alpha_4\alpha_5 \pm \left(\alpha_2\alpha_3\sqrt{\alpha_3^2 + 4\alpha_4\alpha_5} + \alpha_1\alpha_4\sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)}{2\alpha_1\alpha_2\alpha_3 + \alpha_1^2\alpha_4 - 4\alpha_2^2\alpha_5} \text{ and}$$

$$G^* = \frac{\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}}{2\alpha_4}$$

which satisfies the following inequalities:

$$\left(2pQD_1 + 2QTc_pD_1 + QT^2E_hD_1\right) > 2Q + QT^2c_hD_1 + 2dc_1E_TD_1$$

and

$$\left[b\left(2Q + D_1\left(QT^2c_h - 2pQ - 2QTc_p - QT^2E_h + 2dc_1E_T\right)\right)\right] > \frac{(\alpha_3 - 2bdGc_1D_1E_T)^2}{\left(2bdG^2c_1D_1E_T - 2(G\alpha_3 + \alpha_5)\right)}$$

Proof: See Appendix A.1

For a certain taxation policy, both the optimal solution to reduce carbon emission and the total cost can be derived. The results comparing the taxation policy models with and without green investment are illustrated in Proposition 1.

Proposition 1. The green technology investment increases the total profit by

$\frac{1}{T}[c_1R(G) - G] > 0$ under the carbon tax policy, that is,

$$\frac{1}{T}\left[c_1\left(\frac{a\left(\alpha_3\pm\sqrt{\alpha_3^2+4\alpha_4\alpha_5}\right)}{2\alpha_4} - \frac{b\left(\alpha_3\pm\sqrt{\alpha_3^2+4\alpha_4\alpha_5}\right)^2}{4\alpha_4^2}\right) - \frac{\alpha_3\pm\sqrt{\alpha_3^2+4\alpha_4\alpha_5}}{2\alpha_4}\right] > 0$$

and reduce the emission by $[R(G)] > 0$ i.e.,

$$\frac{1}{T}\left[\frac{a\left(\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)}{2\alpha_4} - \frac{b\left(\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)^2}{4\alpha_4^2}\right] > 0$$

Proof: Please see Appendix A.2

It is clear from Proposition 1 that the profit is proportional to the green technology investment, while carbon emission is inversely proportional to green technology investment. This is logical since the green technology investment has reduced carbon emission as well as the carbon tax.

3.4.2. Profit function under cap and trade regulation

The government may set a limit U for the total allowable carbon emissions. Considering the total emissions from the related sources, the retailer can sell the surplus at a rate of C_2. Subsequently, the retailer must buy more allowances from other institutions or invest in green technology when it passes the limit U.

Following the assumptions in Huang et al. (2020), the total carbon emissions cost under the cap and trade regulation is

$$TC_2 = C_2[E(Q,\ G)] - C_2U$$

$$TC_2 = C_2\left[\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right] - C_2U \tag{15}$$

The profit function considering green technology cost under this case is

$$X_2 = SR - HC - PC - TC_2 - G - OC$$

$$= pDT - C_h\frac{DT^2}{2} - C_pDT - C_2\left[\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right] + C_2U - G - OC \tag{16}$$

The total profit per year is

$$TP_2(Q,G) = \frac{X_2}{T}$$

$$TP_2 = \frac{1}{T}\left[pDT - C_h\frac{DT^2}{2} - C_pDT - C_2\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right) + C_2U - G - OC\right] \tag{17}$$

After substituting the value of D into Eq. (17) and simplified, one has:

$$TP_2 = \frac{1}{T}\left[\begin{array}{c} pTD_0 - \frac{D_0C_hT^2}{2} - \frac{C_hT^2mv}{2} - C_pTD_0 - C_pTmv + \frac{T^2(E_hC_2D_0+E_hC_2mv)}{2} + \\ pTmv + C_2U + G\left(pTD_1a - \frac{D_1aC_hT^2}{2} - C_pTD_1a + \frac{aE_hC_2D_1T^2}{2} - C_2a - 1\right) \\ +G^2\left(\frac{C_hT^2}{2}D_1b + C_pTD_1b + C_2b - pTD_1b - \frac{bE_hC_2D_1T^2}{2}\right) \\ -\frac{1}{Q}(C_2E_TdD_0 + C_2E_Tdmv) - \frac{G}{Q}(D_1C_2E_Tda) + \frac{G^2}{Q}D_1C_2E_Tdb \end{array}\right]$$

$$\Rightarrow TP_2 = \frac{1}{T}\left[G\beta_1 + G^2\beta_2 - \frac{G}{Q}\beta_3 + \frac{G^2}{Q}\beta_4 - \frac{1}{Q}\beta_5 + \beta_6\right] \tag{18}$$

where,

$$\beta_1 = pTD_1a - \frac{D_1aC_hT^2}{2} - C_pTD_1a + \frac{aE_hC_2D_1T^2}{2} - C_2a - 1$$

$$\beta_2 = \frac{C_hT^2}{2}D_1b + C_pTD_1b + C_2b - pTD_1b - \frac{bE_hC_2D_1T^2}{2}$$

$$\beta_3 = D_1C_2E_Tda$$
$$\beta_4 = D_1C_2E_Tdb$$
$$\beta_5 = C_2E_TdD_0 + C_2E_Tdmv$$

$$\beta_6 = \left[\begin{array}{c} -C_pTmv + \frac{T^2(E_hC_2D_0+E_hC_2mv)}{2} + C_2U \\ pTD_0 + pTmv - \frac{D_0C_hT^2}{2} - \frac{C_hT^2mv}{2} - C_pTD_0 \end{array}\right]$$

Lemma 2. Under the carbon cap and trade policy, the optimal value of $TP_2(Q,G)$ is concave. The optimal lot size and green technology cost are:

$$Q^* = \frac{\beta_2\beta_3^2 + 4\beta_2\beta_4\beta_5 \pm \left(\beta_2\beta_3\sqrt{\beta_3^2 + 4\beta_4\beta_5} + \beta_1\beta_4\sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_1\beta_2\beta_3 + \beta_1^2\beta_4 - 4\beta_2^2\beta_5} \text{ and}$$

$$G^* = \frac{\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}}{2\beta_4}$$

when the following inequalities are satisfied:

$$QT^2E_hD_1c_2 + 2pQTD_1 > QTD_1(Tc_h + 2c_p) + 2c_2(Q + dE_TD_1)$$

and

$$\left[\begin{array}{c} \left(QT(-2p + Tc_h + 2c_p)D_1 + c_2\left(2Q + D_1\left(-QT^2e_h + 2de_T\right)\right)\right) \\ \left(-d(a-2bG)^2c_2D_1^2e_T - 2b(mv + D_0 + G(a-bG)D_1)\right) \end{array}\right] > 0$$

Proof: Please see Appendix B.1

Proposition 2 discusses the effect of green technology investment under the cap-and-trade policy.

Proposition 2. In the case of carbon trading, green technology investment increases the total profit by $\frac{1}{T}[c_2R(G) - G] > 0$, i.e.,

$$\frac{1}{T}\left[c_2\left(\frac{a\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_4} - \frac{b\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)^2}{4\beta_4^2}\right) - \frac{\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}}{2\beta_4}\right] > 0$$

and reduces the emission by $[R(G)] > 0$ i.e.,

$$\frac{1}{T}\left[\frac{a\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_4} - \frac{b\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)^2}{4\beta_4^2}\right] > 0$$

Proof: Please see Appendix B.2

Proposition 2 confirms that the profit is proportional to the green technology investment, while carbon emission is inversely proportional to the green technology investment under cap and trade policy. This shows green technology investment is profitable because it reduces the cost of carbon trade.

3.4.3. Profit function under limited carbon emission regulation

In this policy, the retailer must comply with a strict carbon emissions limit W. Excessive carbon emissions must be reduced by investing in green technology. The sum of emitted carbons from all related sources minus the reduction of carbons due to green technology investment must be equal to the marginal carbon emission limit W. So, the inventory model under this policy is subject to the constraint:

$$\left[\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right] = W$$

Similar to Huang et al. (2020), the Lagrange multiplier method can be used to obtain the maximum profit and optimal green technology investment. The problem for profit

$$X_3 = \left[pDT - C_h\frac{DT^2}{2} - C_pDT - G - OC - \psi(E(Q,\ G) - W)\right] \tag{19}$$

becomes

Maximize

$$TP_3(G,Q,\psi) = \frac{X_3}{T}$$

$$TP_3(G,Q,\psi)$$

$$= \frac{1}{T}\left[pDT - C_h\frac{DT^2}{2} - C_pDT - G - OC - \psi\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G) - W\right)\right] \tag{20}$$

By substituting the value $Q = DT$ in Eq. (21), the total profit with two variables (green technology cost and ψ) is:

$$TP_3(G,\psi) = \frac{1}{T}\begin{bmatrix} pTD_0 + pTmv - \frac{C_hT^2}{2}D_0 - \frac{C_hT^2}{2}mv - C_pTD_0 - \frac{E_Td}{T} - C_pTmv \\ -G\left[\frac{D_1aC_hT^2}{2} + C_pTD_1a + 1 - pTD_1a\right] \\ +G^2\left[\frac{D_1bC_hT^2}{2} - pTD_1b + C_pTD_1b\right] \\ -\psi\left[\frac{E_Td}{T} + \frac{D_0E_hT^2}{2} + \frac{mvE_hT^2}{2} - W\right] \\ -\psi G\left[a - \frac{E_hT^2aD_1}{2}\right] + \psi G^2\left[b - \frac{D_1bE_hT^2}{2}\right] \end{bmatrix}$$

$$TP_3(G,\psi) = \frac{1}{T}\left[\phi_1 - G\phi_2 + G^2\phi_3 + \psi\phi_4 - \psi G\phi_5 + \psi G^2\phi_6\right] \quad (21)$$

where,

$$\phi_1 = pTD_0 + pTmv - \frac{C_hT^2}{2}D_0 - \frac{C_hT^2}{2}mv - C_pTD_0 - \frac{E_Td}{T} - C_pTmv$$

$$\phi_2 = \frac{D_1aC_hT^2}{2} + C_pTD_1a + 1 - pTD_1a$$

$$\phi_3 = \frac{D_1bC_hT^2}{2} - pTD_1b + C_pTD_1b$$

$$\phi_4 = \frac{D_0E_hT^2}{2} + \frac{mvE_hT^2}{2} - W$$

$$\phi_5 = a - \frac{E_hT^2aD_1}{2}$$

$$\phi_6 = b - \frac{D_1bE_hT^2}{2}$$

Lemma 3. Under a limited carbon emissions policy, the optimal $TP_3(Q,G)$ value is concave. The green technology cost (in terms of ψ) is:

$$G^*(\psi) = \frac{\phi_2 + \psi\phi_5}{2(\phi_3 + 2\psi\phi_6)}$$

where the following inequalities must be satisfied
$\phi_3 < 0$ and $\phi_6 < 0$.

Proof: Please see Appendix C.1

Proposition 3 shows the positive impact of green investment to reduce carbon emission under a limited carbon emission policy.

Proposition 3. For the case with limited carbon emission policy, the green technology investment increases the total profit by $\frac{1}{T}[\psi R(G) - G\ ] > 0$ i.e.,

$$\frac{1}{T}\left[\psi\left(-\frac{b(\phi_2+\psi\phi_5)^2}{4(\phi_3+2\psi\phi_6)^2} + \frac{a(\phi_2+\psi\phi_5)}{2(\phi_3+2\psi\phi_6)}\right) - \frac{\phi_2+\psi\phi_5}{2(\phi_3+2\psi\phi_6)}\right] > 0$$

and reduce the emission by $[R(G)] > 0$ i.e.,

$$\frac{1}{T}\left[\frac{a(\phi_2+\psi\phi_5)}{2(\phi_3+2\psi\phi_6)} - \frac{b(\phi_2+\psi\phi_5)^2}{4(\phi_3+2\psi\phi_6)^2}\right] > 0$$

Proof: Please see Appendix C.2

4. Result

In this section, we apply the developed model to study the behavior of the system. Some numerical examples are provided, followed by a sensitivity analysis.

4.1. Numerical illustrations

The following numerical examples are provided to illustrate the models under different carbon pricing regulations. The optimal results are derived for the considered data set fixed from convenient simulations. The input parameters are stated in Table 2.

Table 2
Input parameters for numerical illustrations.

	Input parameters
Common Inputs (CI)	$a = 5$, $b = 0.5$, $OC = 100$, $v = 20$, $m = 0.3$, $C_0 = 4$, $C_T = 1$, $C_h = 0.7$, $d = 160.9$, $E_T = 0.3$, $E_h = 0.2$, $D_0 = 30$, $p = 15$, and $D_1 = 10$
Case 1: Carbon tax	CI, $C_1 = 15$
Case 2: Cap-and-trade	CI, $C_2 = 12$ and $U = 4$
Case 3: Carbon limit	CI, $W = 10$ and $\psi = 5$

Table 3
Optimum results of models.

Cases	Optimum results			Carbon reduced	
	Q^*	G^*	E^*	TP_i^*	$R^*(G)$
1	264.03	4.990	60.25	902.46	12.50
2	256.65	4.991	58.71	1042.45	12.50
3	243.19	4.992	56.21	1302.41	12.50

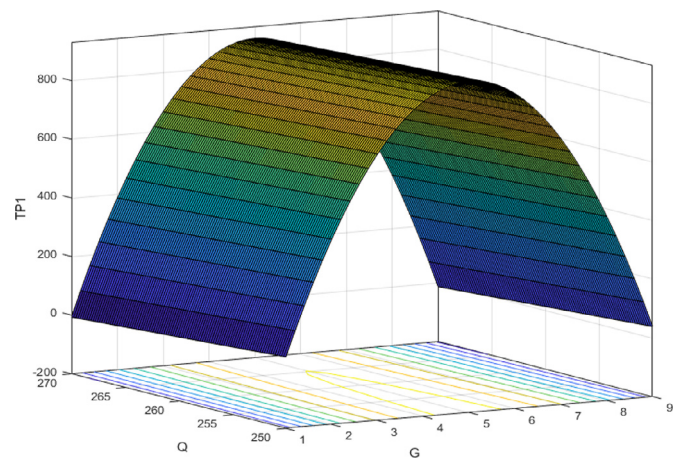

Fig. 2. TP_1 with regard to Q, G.

Our objective is to find the optimal profit, lot size and green investment for each case. The optimum results of the problems:

$$\text{Maximize } TP_i(Q,G) = \frac{X_i}{T}, \text{ for } i = 1,\ 2,\ 3$$

$$\text{subject to } \begin{cases} D > 0,\ G < \frac{a}{b}; i = 1 \\ D, U > 0,\ G < \frac{a}{b}; i = 2 \\ D > 0,\ G < \frac{a}{b} \text{ and } \left[\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right] = W; i = 3 \end{cases}$$

are solved by LINGO 17 software with the unlimited default run time limit, i.e., no limit is imposed on the length of time the solver can run in an Intel(R) Core (TM) i3-7100U CPU@4.40 GHz personal computer. Table 3 represents the results for three cases.

Figs. 2–4 show the concavity of the profit functions for various lot sizes and green technology investments in the three cases.

4.2. Sensitivity analysis

In this section, the sensitivity analysis is carried out based on the numerical examples stated in 4.1. The total profits for all three cases are derived when different parameters vary from -30% to +30%.

4.2.1. Analysis of cost parameters

This study shows the sensitivity analysis for the cost parameters C_p and C_h to the profits. The impact on the profits is investigated for changing purchase cost C_p and holding cost C_h. The results are

Table 4
The impact on the profits for varying C_p.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
C_p	-30%	3.5	1143.96	1283.95	1543.91
	-20%	4.0	1063.46	1203.45	1463.41
	-10%	4.5	982.96	1122.95	1382.91
	0%	5.0	902.46	1042.45	1302.41
	10%	5.5	821.96	961.95	1221.91
	20%	6.0	741.46	881.45	1141.41
	30%	6.5	660.96	800.95	1060.91

Table 5
Changes in profits concerning the variation of C_h.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
C_h	-30%	0.49	930.45	1069.70	1328.50
	-20%	0.56	921.06	1060.55	1319.67
	-10%	0.63	911.73	1051.46	1310.98
	0%	0.70	902.46	1042.45	1302.41
	10%	0.77	893.25	1033.50	1293.95
	20%	0.84	884.09	1024.61	1285.61
	30%	0.91	874.99	1015.78	1277.37

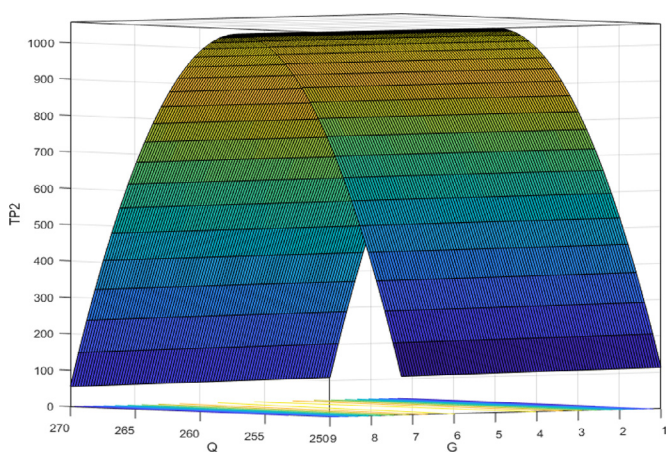

Fig. 3. TP_2 with regard to Q, G.

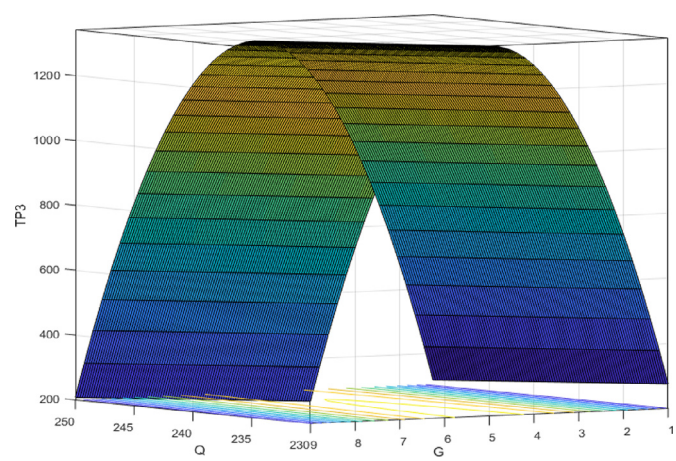

Fig. 4. TP_3 with regard to Q, G.

given in Tables 4 & 5 and their percentage is graphically shown in Figs. 5 & 6 respectively.

Fig. 5 represents the percentage variation in profits of Table 4. The profits for all the cases are changing inversely with the variation of C_p. The purchase cost plays a vital role in the profit function of all three cases. It has the greatest influence on the total profit of the first case, i.e., carbon taxation as C_p has a greater impact on TP_1. For the carbon taxation policy, the variation of C_p from -30% to 30%, the total profit varies from 26.76% to -26.76%.

Fig. 6 represents the percentage variation in profits of Table 5. From Table 5 and Fig. 6, we see that the holding cost has a negative impact on all the total profits. The holding cost influences the total profit of carbon taxation policy, carbon trading policy and limited carbon policy in descending order. They vary slightly from one another.

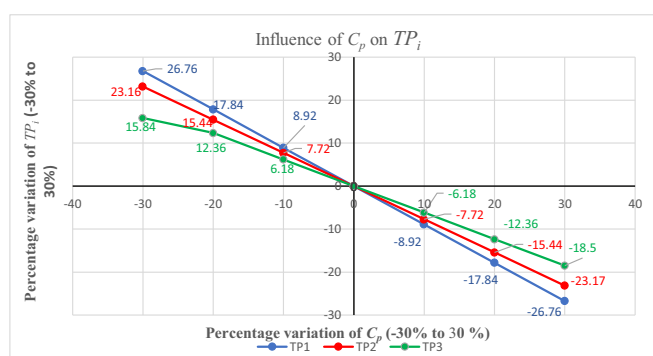

Fig. 5. The relationship between C_p and the total profits.

4.2.2. Analysis of emission parameters

The effects of carbon emission parameters E_T and E_h on the profits are investigated. The results for changing of E_T and E_h are given in Tables 6 & 7 and their percentage variation is graphically illustrated in Figs. 7 & 8 respectively.

Fig. 7 illustrates the percentage of variation in profits. For the variation of E_T from -30% to 30%, total profit varies from 10.12% to -8.14% for carbon taxation policy, varies from 7.45% to -5.95% for carbon trading policy and varies from 2.74% to -2.25% for limited emission policy. Hence, E_T has a greater impact on TP_1 among other parameters.

Table 6
Changes in profits concerning the variation of E_T.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
E_T	-30%	0.21	993.82	1120.06	1338.07
	-20%	0.24	960.69	1091.85	1325.20
	-10%	0.27	930.42	1066.14	1313.37
	0%	0.30	902.46	1042.45	1302.41
	10%	0.33	876.43	1020.43	1292.16
	20%	0.36	852.02	999.81	1282.53
	30%	0.39	829.01	980.40	1273.43

Table 7
Changes in profits on the variation of E_h.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
E_h	-30%	0.14	1087.32	1138.65	1340.05
	-20%	0.16	1057.64	1105.67	1327.23
	-10%	0.18	1030.57	1073.64	1314.69
	0%	0.20	902.46	1042.45	1302.41
	10%	0.22	863.36	1012.02	1290.36
	20%	0.24	825.19	982.27	1278.54
	30%	0.26	787.87	953.16	1266.92

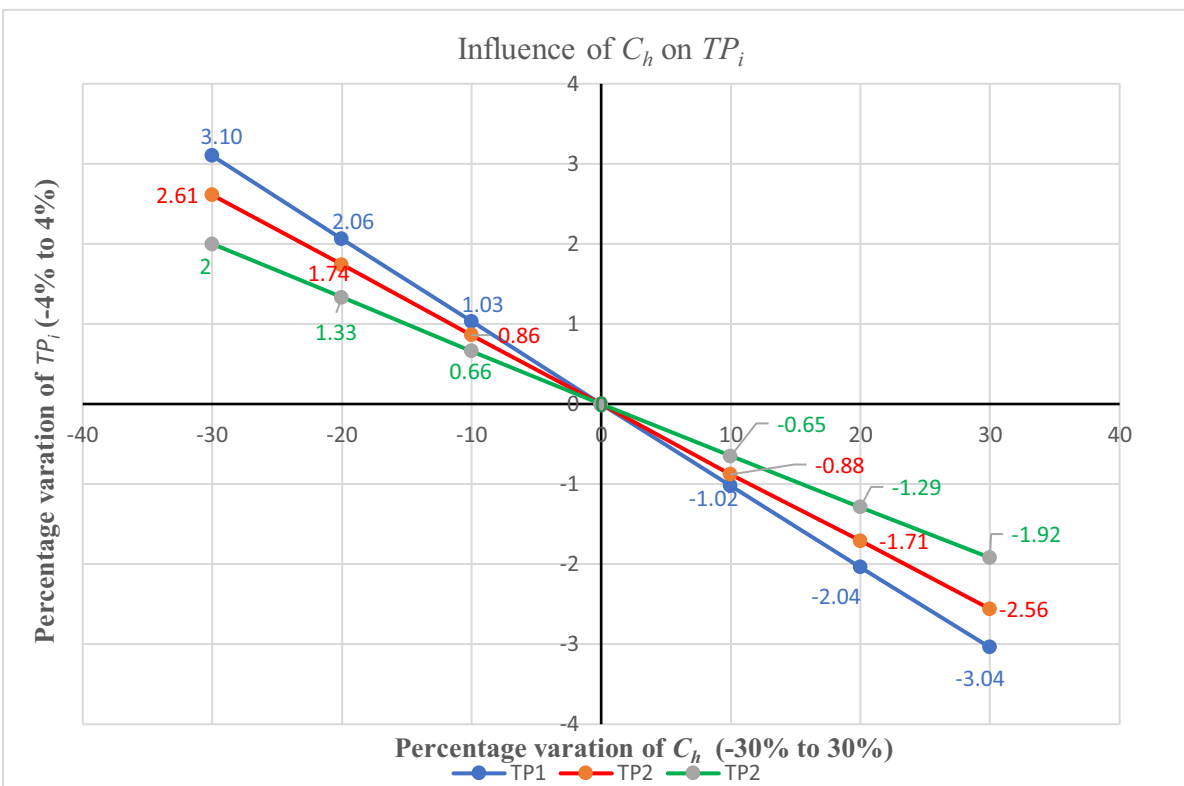

Fig. 6. The relationship between C_h and the total profits.

Fig. 8 illustrates the percentage of variation in profits. Emission parameter E_h has a high negative impact on the total profit for carbon taxation policy. In the first case, when E_h varies from -30% to 30%, the total profit varies from 20.48% to -12.70%. Increasing E_h also decreases in total profits of carbon trading policy and limited carbon policy.

4.2.3. Analysis of green investment factors

This section investigates how the profits are influenced by the efficiency factor of carbon reduction (a), and the emission factor of green technology (b). The results are given in Tables 8 & 9 and their percentage variation is graphically illustrated in Figs. 9 & 10.

Fig. 9 illustrates the percentage of variation in profits. From Table 8 and Fig. 9, we see that the reduction fraction is identical and positively impacts the total profits for all the cases. The first case has the highest variation of the total profit from -43.67% to 59.27%. The impact on the total profit decreases for the second case and the third case.

Fig. 10 illustrates the percentage of variation in profits. From the figure, the third case (limited carbon policy) has the highest neg-

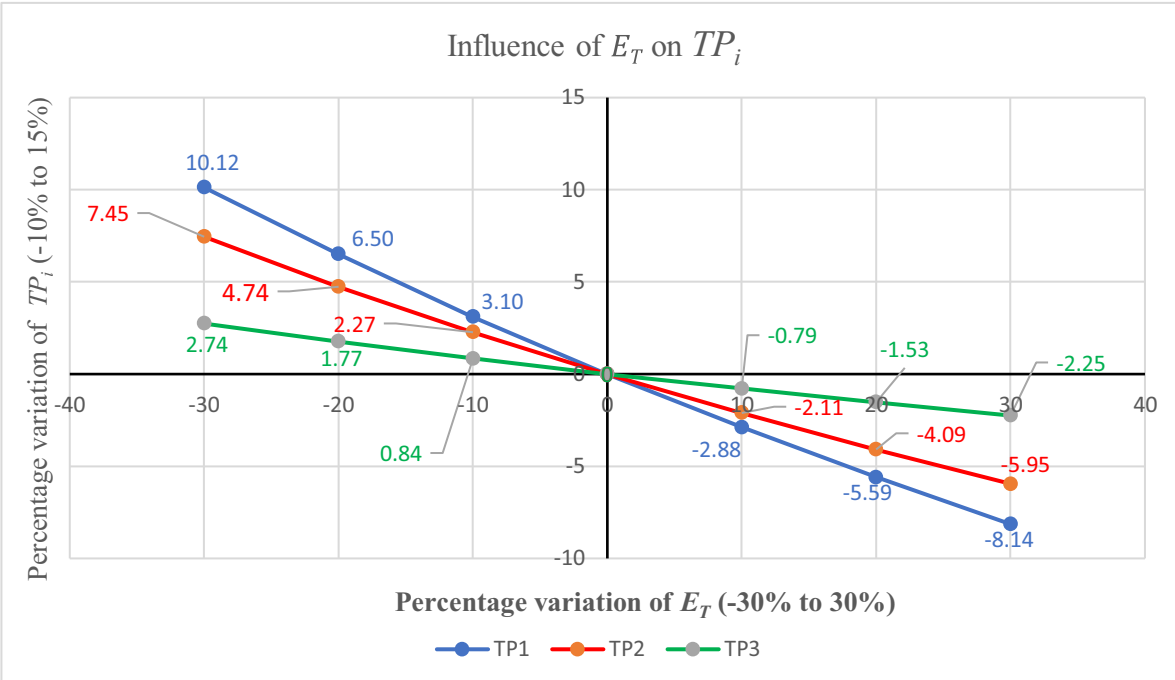

Fig. 7. The relationship between E_T and the total profit.

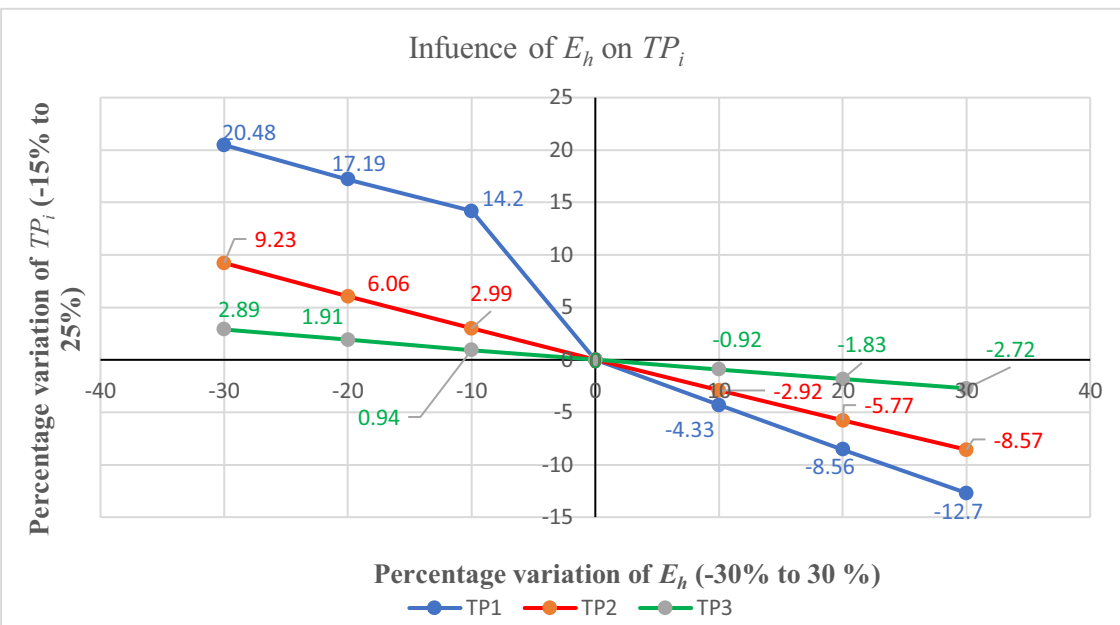

Fig. 8. The relationship between E_h and the total profit.

Table 8
Changes in profits for varying *a*.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
a	-30%	3.5	508.39	606.10	768.98
	-20%	4.0	624.05	734.30	925.72
	-10%	4.5	755.41	879.75	1103.53
	0%	5.0	902.46	1042.45	1302.41
	10%	5.5	1065.17	1222.38	1522.35
	20%	6.0	1243.53	1419.54	1763.33
	30%	6.5	1437.52	1633.92	2025.35

Table 9
Changes in profits for varying *b*.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
b	-30%	0.35	1233.94	1409.00	1750.57
	-20%	0.40	1095.78	1256.25	1563.81
	-10%	0.45	988.36	1137.46	1418.58
	0%	0.50	902.46	1042.45	1302.41
	10%	0.55	832.21	964.73	1207.38
	20%	0.60	773.70	899.97	1128.21
	30%	0.65	724.21	845.19	1061.22

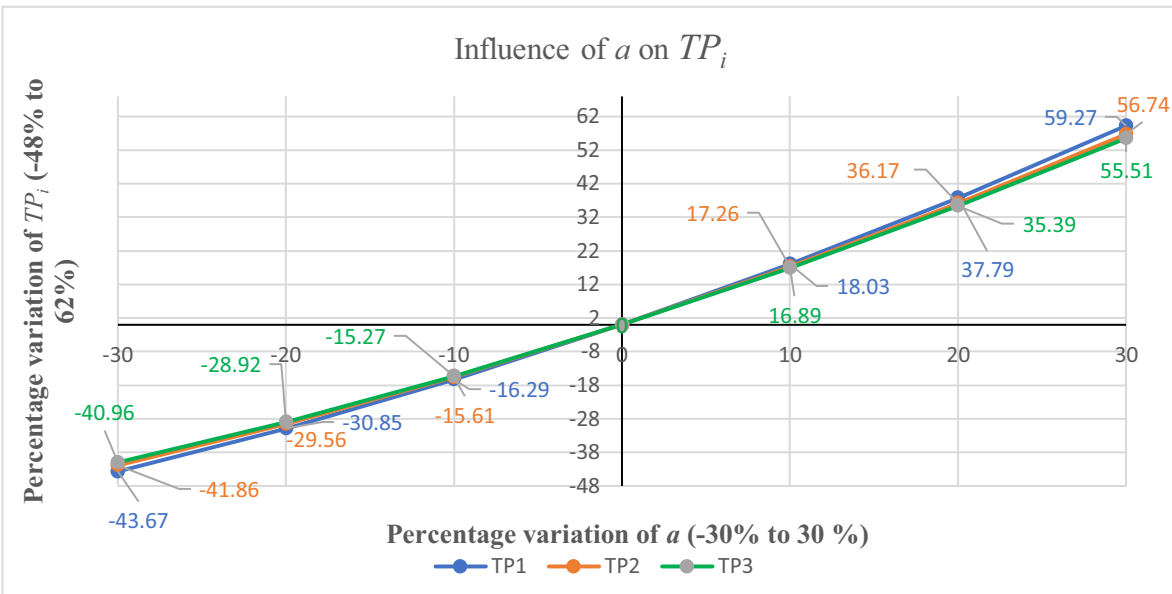

Fig. 9. The relationship between carbon reduction factor *a* and the total profit.

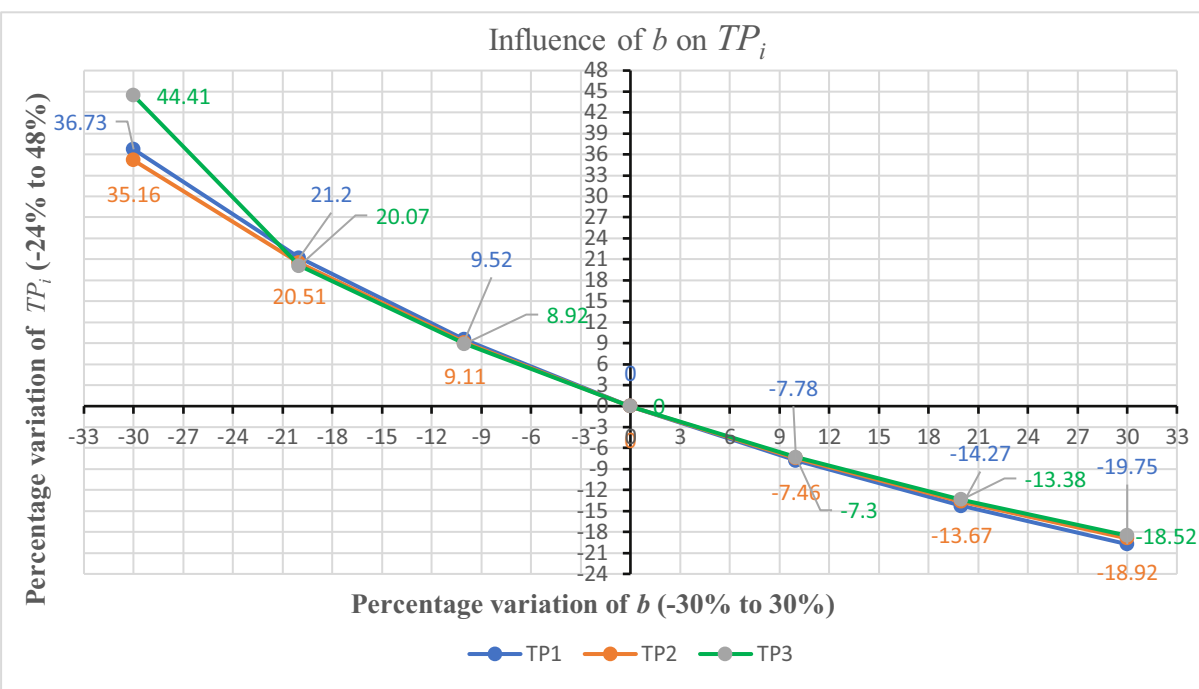

Fig. 10. The relationship between the emission factor *b* and the total profit.

Table 10
Changes in profit TP_1 for varying C_1.

Parameter	Percentage	Variation of parameters	Variation in TP_1^*
C_1	-30%	10.5	1067.83
	-20%	12.0	1012.70
	-10%	13.5	957.58
	0%	15.0	902.46
	10%	16.5	847.35
	20%	18.0	792.25
	30%	19.5	737.15

Table 11
Changes in profit TP_2 for varying U.

Parameter	Percentage	Variation of parameters	Variation in TP_2^*
U	-30%	2.8	1033.45
	-20%	3.2	1036.44
	-10%	3.6	1039.44
	0%	4.0	1042.45
	10%	4. 4	1045.46
	20%	4.8	1048.49
	30%	19.5	737.15

Table 12
Changes in profit TP_2 for varying C_2.

Parameter	Percentage	Variation of parameters	Variation in TP_2^*
C_2	-30%	8. 4	1166.03
	-20%	9.6	1124.83
	-10%	10.8	1083.64
	0%	12.0	1042.45
	10%	13.2	1001.27
	20%	14. 4	960.09
	30%	15.6	918.91

ative impact of the emission factor of green technology b on the total profit. The highest total profit variation is 44.41% to -18.52% when b varies from -30% to 30%. The impact decreases for the first case and second case.

4.2.4. Analysis of the carbon tax C_1

The effects of the carbon tax on unit carbon emissions are related to carbon taxation policy only. Table 10 presents the variations of total profit numerically. One can see that more tax in the carbon taxation policy results in less profit.

4.2.5. Analysis for carbon emission cap U

The effects of parameter U is only related to the model under cap and trade policy. Table 11 shows the sensitivity analysis of U on total profit (TP_2^*). Under a higher limit of U, the total profit will be maximum. Cap and trade policy with a higher limit of U gives a chance to trade more carbons, although the percentage variation of TP is quite small.

4.2.6. Analysis of carbon trading price C_2

The effects of carbon trading price C_2 related to carbon cap and trade policy only. Table 12 shows the effect of varying C_2 on the total profit. The table shows that the increasing value of carbon trading price C_2 has a negative impact on the total profit. At a certain time, the managers may have to buy credit from another company for maintaining carbon cap and trade regulation. At another time, he may sell the credit to another company. In that situation, C_2 will have a reverse impact, i.e., the total profit will be greater with a higher value of C_2.

4.2.7. Analysis of carbon emission limit W

For a limited carbon emission policy, Table 13 presents the sensitivity analysis of the limit W on total profit. Table 13 indicates

Table 13
Changes in profit TP_3 for varying W.

Parameter	Percentage	Variation of parameters	Variation in TP_3^*
W	-30%	7	1292.56
	-20%	8	1295.82
	-10%	9	1299.11
	0%	10	1302.41
	10%	11	1305.73
	20%	12	1309.06
	-30%	7	1292.56

that W has a positive impact on the total profit TP_3. For the higher limit of W, managers gain more space to execute operations and earn more.

4.2.8. Analysis of parameter v

Table 14 and Fig. 11 show the relationship between the promotion level v and the total inventory profit. The impact of v on the total profit is highest for the third case, varying from −1.11% to +1.11%. The variation of the total profit for the first case and second case are similar (the TP_1 line and TP_2 line almost coincide with each other). Since v is proportional to the profit due to higher demand, increasing green investment and its promotion can sustain a profitable inventory management system.

5. Discussion

This study investigates economic order quantity decisions in which the demand function depends on the impact of green technology investment and its promotion. Three models are developed under a carbon tax, cap-and-trade, and limited carbon emission regulations. From the numerical examples and sensitivity analysis, the following insights can be derived:

a) For different carbon emission regulations, companies can adjust inventory levels and green technology investment to optimize the total profits. From the theoretical derivation of Lemmas 1–3, it is confirmed that the total profit is a concave function.
b) Propositions 1–3 ensure that investment in green technology is worth undertaking due to increasing benefits. The result confirms the study by Toptal et al. (2014). In addition, the changes in efficiency factors and emission factors from green technology do not affect the corresponding response to different carbon emission regulations.
c) For different carbon emission regulations, there are considerable differences in the effect of emission levels on the total profits from the transportation and storage activities. Changes in emission levels have the greatest effect on the carbon tax system and the least on limited carbon policy. In other words, reducing emission levels will result in greater benefits in the carbon tax system. The company's desire to reduce emissions such as investing in green technology, using renewable energy, or redesigning the logistic network is greater with a carbon tax system.
d) Since the model we have developed takes into account the positive impact of promotion on demand, the total profit is proportional to the promotion level. From the numerical analysis, it is found that this effect is similar in all the carbon emission regulations.
e) Regarding the whole decision model, managers must pay more attention to carbon reduction efficiency, the emission factor of green technology, carbon emission parameters from transportation logistics and storage, and the purchase cost since the changes in these parameters have significant impacts on the total profit.

Table 14
Changes in profits for varying v.

Parameter	Percentage	Variation of parameters	Variation in TP_i^*		
			TP_1^*	TP_2^*	TP_3^*
v	-30%	14	892.93	1031.45	1287.90
	-20%	16	896.11	1035.11	1292.74
	-10%	18	899.29	1038.78	1297.57
	0%	20	902.46	1042.45	1302.41
	10%	22	905.64	1046.12	1307.24
	20%	24	908.81	1049.78	1312.08
	30%	26	911.99	1053.45	1316.91

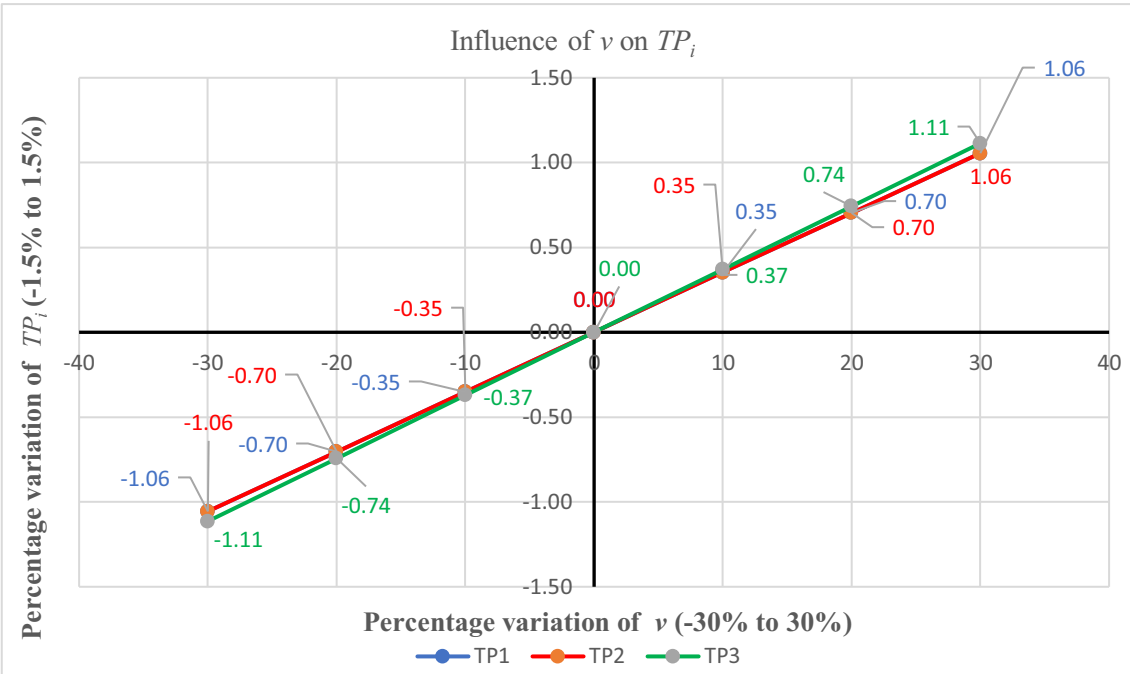

Fig. 11. The relationship between promotion level and the total profit.

f) Carbon price significantly affects the total profit in the carbon tax system, whereas the carbon cap on cap-and-trade and strict carbon limit policies are less significant to the total profit. Hence, the government should carefully decide its policy and implementation.

6. Conclusions

This paper investigates a company's efforts to reduce carbon emissions through inventory planning and technology investment under carbon emission regulations: carbon tax, cap-and-trade, and carbon limit. Optimal decisions regarding the order size and the amount of investment per cycle are made while maintaining the company's maximum profit. This study considers increasing customer demand as a result of the company's green technology investment and the promotion of its green performance. The optimal total profit of the inventory model under all the carbon emission regulations is concave with lot size per cycle and green technology investment as the decision variables.

This study produces a decision method on the optimum green technology investment and provides a new insight on low carbon inventory management. The results show that the varying efficiency factors and emission factors from green technology do not affect the carbon emission regulations. Furthermore, this study reveals that the level of emissions from transporting and storing supplies significantly affects the total profit in the carbon tax system. Therefore, under the carbon tax system, reducing the level of emissions will result in greater benefits. Moreover, because the level of the carbon tax significantly affects the total profit, the government should carefully specify the tax level. However, the sensitivity analysis shows that the carbon cap in cap-and-trade and strict carbon limit policies are less significant on the total profits.

Our model assumes a green technology investment and promotion-dependent demand in a linear function. Future research can be done to consider other nonlinear functions. Our research is limited to the study of a buyer's inventory model. Further works can be done for a production-inventory model of a manufacturer or in a supply chain system.

Declaration of Competing Interest

None.

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Appendix A.1

Proof of Lemma 1

To prove the concavity of the objective function, the following procedure is used:

Step 1: Find $\frac{\partial TP_1(Q,G)}{\partial Q}$, $\frac{\partial TP_1(Q,G)}{\partial Q}$

Differentiating Eq. (14) with respect to Q, and set the result equal to zero. One has:

$$\frac{\partial TP_1(Q,G)}{\partial Q} = \frac{\partial}{\partial Q}\left[\frac{1}{T}\left(G\alpha_1 + G^2\alpha_2 - \frac{G}{Q}\alpha_3 + \frac{G^2}{Q}\alpha_4 - \frac{1}{Q}\alpha_5 + \alpha_6\right)\right]$$

$$\frac{\partial TP_1(Q,G)}{\partial Q} = \frac{\frac{G\alpha_3}{Q^2} - \frac{G^2\alpha_4}{Q^2} + \frac{\alpha_5}{Q^2}}{T} = 0 \tag{A.1.1}$$

Differentiating Eq. (14) with respect to G, and set the result equal to zero. One has:

$$\frac{\partial TP_1(Q,G)}{\partial G} = \frac{\partial}{\partial G}\left[\frac{1}{T}\left(G\alpha_1 + G^2\alpha_2 - \frac{G}{Q}\alpha_3 + \frac{G^2}{Q}\alpha_4 - \frac{1}{Q}\alpha_5 + \alpha_6\right)\right]$$

$$\frac{\partial TP_1(Q,G)}{\partial G} = \frac{\alpha_1 + 2G\alpha_2 - \frac{\alpha_3}{Q} + \frac{2G\alpha_4}{Q}}{T} = 0 \tag{A.1.2}$$

Step 2: Derive Q^*, G^* from **step 1**

Solving equations (A.1.1) and (A.1.2). One has:

$$Q^* = \frac{\alpha_2\alpha_3^2 + 4\alpha_2\alpha_4\alpha_5 \pm \left(\alpha_2\alpha_3\sqrt{\alpha_3^2 + 4\alpha_4\alpha_5} + \alpha_1\alpha_4\sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)}{2\alpha_1\alpha_2\alpha_3 + \alpha_1^2\alpha_4 - 4\alpha_2^2\alpha_5}$$

$$G^* = \frac{\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}}{2\alpha_4}$$

Thus, the optimal point is:

$$(Q^*, G^*) \equiv \begin{bmatrix} \frac{\alpha_2\alpha_3^2 + 4\alpha_2\alpha_4\alpha_5 \pm \left(\alpha_2\alpha_3\sqrt{\alpha_3^2 + 4\alpha_4\alpha_5} + \alpha_1\alpha_4\sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)}{2\alpha_1\alpha_2\alpha_3 + \alpha_1^2\alpha_4 - 4\alpha_2^2\alpha_5}, \\ \frac{\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}}{2\alpha_4} \end{bmatrix}$$

Step 3: Find $\frac{\partial^2 TP_1(Q,G)}{\partial Q^2}$, $\frac{\partial^2 TP_1(Q,G)}{\partial G^2}$ and $\frac{\partial^2 TP_1(Q,G)}{\partial Q \partial G}$

Differentiate (A.1.1) with respect to Q,

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q^2} = \frac{\partial}{\partial Q}\left[\frac{\frac{G\alpha_3}{Q^2} - \frac{G^2\alpha_4}{Q^2} + \frac{\alpha_5}{Q^2}}{T}\right]$$

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q^2} = \frac{-\frac{2G\alpha_3}{Q^3} + \frac{2G^2\alpha_4}{Q^3} - \frac{2\alpha_5}{Q^3}}{T} \tag{A.1.3}$$

Again differentiate (A.1.2) with respect to G,

$$\frac{\partial^2 TP_1(Q,G)}{\partial G^2} = \frac{\partial}{\partial G}\left[\frac{\alpha_1 + 2G\alpha_2 - \frac{\alpha_3}{Q} + \frac{2G\alpha_4}{Q}}{T}\right]$$

$$\frac{\partial^2 TP_1(Q,G)}{\partial G^2} = \frac{2\alpha_2 + \frac{2\alpha_4}{Q}}{T} \tag{A.1.4}$$

Now differentiate (A.1.1) with respect to G,

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q \partial G} = \frac{\partial}{\partial G}\left[\frac{\frac{G\alpha_3}{Q^2} - \frac{G^2\alpha_4}{Q^2} + \frac{\alpha_5}{Q^2}}{T}\right]$$

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q \partial G} = \frac{\frac{\alpha_3}{Q^2} - \frac{2G\alpha_4}{Q^2}}{T} \tag{A.1.5}$$

Step 4: Proof the optimality condition of the objective function.

To ensure that the optimal profit function is concave, the following conditions must be satisfied:

$\frac{\partial^2 TP_1(Q,G)}{\partial Q^2} < 0$, $\frac{\partial^2 TP_1(Q,G)}{\partial G^2} < 0$ and

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q^2} \times \frac{\partial^2 TP_1(Q,G)}{\partial G^2} - \left(\frac{\partial^2 TP_1(Q,G)}{\partial Q \partial G}\right)^2 > 0$$

Now we have,

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q^2} = \frac{-\frac{2G\alpha_3}{Q^3} + \frac{2G^2\alpha_4}{Q^3} - \frac{2\alpha_5}{Q^3}}{T}$$

Simplifying, we have

$$-\frac{2dC_1(mv + D_0 + G(a - bG)D_1)E_T}{Q^3 T}$$

which will be less than zero if $a - bG$ is positive; i.e. for $a > bG$ (our assumptions).

Again, $\frac{\partial^2 TP_1(Q,G)}{\partial G^2} = \frac{2\alpha_2 + \frac{2\alpha_4}{Q}}{T}$

Simplifying, we have

$$\frac{b\left(2Q + QT^2c_hD_1 + 2dc_1E_TD_1 - \left(2pQD_1 + 2QTc_pD_1 + QT^2E_hD_1\right)\right)}{QT}$$

which is less than zero if

$$\left(2pQD_1 + 2QTc_pD_1 + QT^2E_hD_1\right) > 2Q + QT^2c_hD_1 + 2dc_1E_TD_1 \tag{A.1.6}$$

and

$$\frac{\partial^2 TP_1(Q,G)}{\partial Q^2} \times \frac{\partial^2 TP_1(Q,G)}{\partial G^2} - \left(\frac{\partial^2 TP_1(Q,G)}{\partial Q \partial G}\right)^2$$

$$= \left[\left(\frac{-\frac{2G\alpha_3}{Q^3} + \frac{2G^2\alpha_4}{Q^3} - \frac{2\alpha_5}{Q^3}}{T}\right) \times \left(\frac{2\alpha_2 + \frac{2\alpha_4}{Q}}{T}\right) - \left(\frac{\frac{\alpha_3}{Q^2} - \frac{2G\alpha_4}{Q^2}}{T}\right)^2\right]$$

$$\frac{1}{Q^4T^2}\begin{bmatrix} b\left(2Q + D_1\left(-2pQ + QT^2c_h - 2QTc_p - QT^2E_h + 2dc_1E_T\right)\right) \\ \left(2bdG^2c_1D_1E_T - 2(G\alpha_3 + \alpha_5)\right) \\ -(\alpha_3 - 2bdGc_1D_1E_T)^2 \end{bmatrix} \tag{A.1.7}$$

Since $\frac{1}{Q^4T^2}$ is positive, (A.1.7) is positive if

$$\left[b\left(\begin{matrix}\left(2Q + D_1\left(QT^2c_h - 2pQ - 2QTc_p - QT^2E_h + 2dc_1E_T\right)\right) \\ \left(2bdG^2c_1D_1E_T - 2(G\alpha_3 + \alpha_5)\right) \\ -(\alpha_3 - 2bdGc_1D_1E_T)^2\end{matrix}\right)\right] > 0$$

$$\left[b\left(2Q + D_1\left(QT^2c_h - 2pQ - 2QTc_p - QT^2E_h + 2dc_1E_T\right)\right)\right]$$
$$> \frac{(\alpha_3 - 2bdGc_1D_1E_T)^2}{\left(2bdG^2c_1D_1E_T - 2(G\alpha_3 + \alpha_5)\right)} \tag{A.1.8}$$

Finally, if conditions (A.1.6) and (A.1.8) are satisfied, the total profit is always concave.

Proposition 1 discusses the usefulness of investing in carbon emission reduction.

Appendix A.2

Proof of Proposition 1:

For,

$$TP_1(Q,\ G) - TP_1(Q,\ 0) =$$
$$\left[\frac{1}{T}\left(pDT - C_h\left(\frac{DT^2}{2}\right) - C_pDT - C_1\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right) - G - OC\right)\right.$$
$$\left. - \frac{1}{T}\left[pDT - C_h\left(\frac{DT^2}{2}\right) - C_pDT - C_1\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2}\right) - OC\right]\right]$$
$$= \frac{1}{T}[c_1R(G) - G]$$

$$= \frac{1}{T}[c_1(aG - bG^2) - G] \tag{A.2.1}$$

At the optimal point $(Q^*,\ G^*)$,

$$TP_1(Q^*,\ G^*) - TP_1(Q^*,\ 0) = \frac{1}{T}\left[\begin{array}{c} \left(\begin{array}{c} -\frac{\left\{adc_1D_1E_T \pm \sqrt{4bd^2c_1^2(mv+D_0)D_1{E_T}^2 + a^2d^2c_1^2D_1^2{E_T}^2}\right\}^2}{4bd^2c_1^2D_1^2{E_T}^2} \\ +\frac{a\left\{adc_1D_1E_T \pm \sqrt{4bd^2c_1^2(mv+D_0)D_1{E_T}^2 + a^2d^2c_1^2D_1^2{E_T}^2}\right\}}{2bdc_1D_1E_T} \end{array}\right) \\ -\frac{adc_1D_1E_T \pm \sqrt{4bd^2c_1^2(mv+D_0)D_1{E_T}^2 + a^2d^2c_1^2D_1^2{E_T}^2}}{2bdc_1D_1E_T} \end{array}\right]$$

Simplifying, we have

$$TP_1(Q^*,\ G^*) - TP_1(Q^*,\ 0) = \frac{1}{T}\left[\begin{array}{c} ((adc_1D_1E_T \pm \sqrt{d^2c_1^2D_1(4bmv + 4bD_0 + a^2D_1)E_T^2}) + \\ 2d(1 - ac_1)D_1E_T)(\frac{1}{4bd^2c_1D_1^2E_T^2} \\ \pm\sqrt{d^2c_1^2D_1(4bmv + 4bD_0 + a^2D_1)E_T^2} - adc_1D_1\mathrm{e}E_T) \end{array}\right] \tag{A.2.2}$$

From the values $\alpha_3 = D_1C_1E_Tda$, $\alpha_4 = D_1C_1E_Tdb$, $\alpha_5 = C_1E_TdD_0 + C_1E_Tdmv$ in equation (A.2.2), we have

$$TP_1(Q^*,\ G^*) - TP_1(Q^*,\ 0) = \frac{1}{T}\left[c_1\left(\frac{a\left(\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)}{2\alpha_4} - \frac{b\left(\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)^2}{4\alpha_4^2}\right) - \frac{\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}}{2\alpha_4}\right] > 0$$

Thus,

$$\begin{aligned} &E_1(Q,\ 0) - E_1(Q,\ G) \\ &= \left[\tfrac{1}{T}\left(\tfrac{D}{Q}E_Td + E_h\tfrac{DT^2}{2}\right) - \tfrac{1}{T}\left(\tfrac{D}{Q}E_Td + E_h\tfrac{DT^2}{2} - R(G)\right)\right] \\ &= \frac{1}{T}[R(G)] \\ &= \frac{1}{T}\left[aG - bG^2\right] \end{aligned} \tag{A.2.3}$$

At the optimal point $(Q^*,\ G^*)$,

$$E_1(Q^*,\ 0) - E_1(Q^*,\ G^*) = \frac{1}{T}\left[\begin{array}{c} \frac{a\left(adc_1D_1E_T \pm \sqrt{4bd^2c_1^2(mv+D_0)D_1{E_T}^2 + a^2d^2c_1^2D_1^2{E_T}^2}\right)}{2bdc_1D_1E_T} \\ -\frac{\left(adc_1D_1E_T \pm \sqrt{4bd^2c_1^2(mv+D_0)D_1{E_T}^2 + a^2d^2c_1^2D_1^2{E_T}^2}\right)^2}{4bd^2c_1^2D_1^2{E_T}^2} \end{array}\right]$$

Simplifying, we have

$$E_1(Q^*,\ 0) - E_1(Q^*,\ G^*) = \frac{1}{T}\left[\begin{array}{c} \frac{1}{4bd^2c_1^2D_1^2E_T^2} \\ \left(\pm\sqrt{d^2c_1^2D_1E_T^2\left(4bmv + 4bD_0 + a^2D_1\right)} - adc_1D_1E_T\right) \\ \left(\left(adc_1D_1E_T \pm \sqrt{d^2c_1^2D_1\left(4bmv + 4bD_0 + a^2D_1\right)E_T^2}\right) \\ -2adc_1D_1E_T\right) \end{array}\right] \tag{A.2.4}$$

From the values α_3, α_4 and α_5 in equation (A.2.4) we have,

$$E_1(Q^*,\ 0) - E_1(Q^*,\ G^*) = \frac{1}{T}\left[\frac{a\left(\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)}{2\alpha_4} - \frac{b\left(\alpha_3 \pm \sqrt{\alpha_3^2 + 4\alpha_4\alpha_5}\right)^2}{4\alpha_4^2}\right] > 0$$

Therefore, investment in green technology is effective under carbon taxation.

Appendix B.1

Proof of Lemma 2

The following procedure is used to prove the concavity of the function:

Step 1: Find $\frac{\partial TP_2(Q,G)}{\partial Q}$, $\frac{\partial TP_2(Q,G)}{\partial G}$

Differentiate Eq. (18) with respect to Q, and set the result equal to zero, one has:

$$\frac{\partial TP_2(Q,G)}{\partial Q} = \frac{\partial}{\partial Q}\left[\frac{1}{T}\left(G\beta_1 + G^2\beta_2 - \frac{G}{Q}\beta_3 + \frac{G^2}{Q}\beta_4 - \frac{1}{Q}\beta_5 + \beta_6\right)\right]$$

$$\frac{\partial TP_2(Q,G)}{\partial Q} = \frac{\frac{G\beta_3}{Q^2} - \frac{G^2\beta_4}{Q^2} + \frac{\beta_5}{Q^2}}{T} = 0 \tag{B.1.1}$$

Differentiate Eq. (18) with respect to G, and set the result equal to zero, one has:

$$\frac{\partial TP_2(Q,G)}{\partial G} = \frac{\partial}{\partial G}\left[\frac{1}{T}\left(G\beta_1 + G^2\beta_2 - \frac{G}{Q}\beta_3 + \frac{G^2}{Q}\beta_4 - \frac{1}{Q}\beta_5 + \beta_6\right)\right]$$

$$\frac{\partial TP_2(Q,G)}{\partial G} = \frac{\beta_1 + 2G\beta_2 - \frac{\beta_3}{Q} + \frac{2G\beta_4}{Q}}{T} = 0 \tag{B.1.2}$$

Step 2: Find Q^*, G^* from **step 1**

Solving the equations (B.1.1) and (B.1.2), one has:

$$Q^* = \frac{\beta_2\beta_3^2 + 4\beta_2\beta_4\beta_5 \pm \left(\beta_2\beta_3\sqrt{\beta_3^2 + 4\beta_4\beta_5} + \beta_1\beta_4\sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_1\beta_2\beta_3 + \beta_1^2\beta_4 - 4\beta_2^2\beta_5}$$

$$G^* = \frac{\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}}{2\beta_4}$$

Thus, the optimal point is derived.

$$(Q^*, G^*) \equiv \left[\begin{array}{c} \frac{\beta_2\beta_3^2 + 4\beta_2\beta_4\beta_5 \pm \left(\beta_2\beta_3\sqrt{\beta_3^2 + 4\beta_4\beta_5} + \beta_1\beta_4\sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_1\beta_2\beta_3 + \beta_1^2\beta_4 - 4\beta_2^2\beta_5}, \\ \frac{\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}}{2\beta_4} \end{array}\right]$$

Step 3: Find $\frac{\partial^2 TP_2(Q,G)}{\partial Q^2}$, $\frac{\partial^2 TP_2(Q,G)}{\partial G^2}$ and $\frac{\partial^2 TP_2(Q,G)}{\partial Q\partial G}$

Differentiate (B.1.1) with respect to Q,

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q^2} = \frac{\partial}{\partial Q}\left(\frac{\frac{G\beta_3}{Q^2} - \frac{G^2\beta_4}{Q^2} + \frac{\beta_5}{Q^2}}{T}\right)$$

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q^2} = \frac{-\frac{2G\beta_3}{Q^3} + \frac{2G^2\beta_4}{Q^3} - \frac{2\beta_5}{Q^3}}{T} \tag{B.1.3}$$

Again differentiate (B.1.2) with respect to G,

$$\frac{\partial^2 TP_2(Q,G)}{\partial G^2} = \frac{\partial}{\partial G}\left(\frac{\beta_1 + 2G\beta_2 - \frac{\beta_3}{Q} + \frac{2G\beta_4}{Q}}{T}\right)$$

$$\frac{\partial^2 TP_2(Q,G)}{\partial G^2} = \frac{2\beta_2 + \frac{2\beta_4}{Q}}{T} \tag{B.1.4}$$

Differentiate (B.1.1) with respect to G,

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q\partial G} = \frac{\partial}{\partial G}\left(\frac{\frac{G\beta_3}{Q^2} - \frac{G^2\beta_4}{Q^2} + \frac{\beta_5}{Q^2}}{T}\right)$$

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q\partial G} = \frac{\frac{\beta_3}{Q^2} - \frac{2G\beta_4}{Q^2}}{T} \tag{B.1.5}$$

Step 4: The following conditions for optimality of objective function is proved.

$\frac{\partial^2 TP_2(Q,G)}{\partial Q^2} < 0$, $\frac{\partial^2 TP_2(Q,G)}{\partial G^2} < 0$ and

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q^2} \times \frac{\partial^2 TP_2(Q,G)}{\partial G^2} - \left(\frac{\partial^2 TP_2(Q,G)}{\partial Q \partial G}\right)^2 > 0$$

One has,

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q^2} = \frac{-\frac{2G\beta_3}{Q^3} + \frac{2G^2\beta_4}{Q^3} - \frac{2\beta_5}{Q^3}}{T}$$

Simplifying, one has

$$-\frac{2dc_2E_T(mv + D_0 + GD_1(a - bG))}{Q^3T}$$

which will be less than zero if $a - bG$ is positive i.e. $a > bG$ (our assumption).

Again, $\frac{\partial^2 TP_2(Q,G)}{\partial G^2} = \frac{2\beta_2 + \frac{2\beta_4}{Q}}{T}$

Simplifying, one has

$$\frac{b\left[QTD_1(Tc_h + 2c_p) + 2c_2(Q + dE_TD_1) - \left(QT^2E_hD_1c_2 + 2pQTD_1\right)\right]}{QT}$$

which will be less than zero if

$$QT^2E_hD_1c_2 + 2pQTD_1 > \ QTD_1(Tc_h + 2c_p) + 2c_2(Q + dE_TD_1) \tag{B.1.6}$$

and

$$\frac{\partial^2 TP_2(Q,G)}{\partial Q^2} \times \frac{\partial^2 TP_2(Q,G)}{\partial G^2} - \left(\frac{\partial^2 TP_2(Q,G)}{\partial Q \partial G}\right)^2$$
$$= \left[\left(\frac{-\frac{2G\beta_3}{Q^3} + \frac{2G^2\beta_4}{Q^3} - \frac{2\beta_5}{Q^3}}{T}\right) \times \left(\frac{2\beta_2 + \frac{2\beta_4}{Q}}{T}\right) - \left(\frac{\frac{\beta_3}{Q^2} - \frac{2G\beta_4}{Q^2}}{T}\right)^2\right]$$

Simplifying, one has

$$\frac{dc_2e_T}{Q^4T^2}\begin{bmatrix} \left(-d\left(a - 2bG\right)^2c_2D_1^2e_T - 2b\left(mv + D_0 + G\left(a - bG\right)D_1\right)\right. \\ \left(QT\left(-2p + Tc_h + 2c_p\right)D1 + \right. \\ \left. c_2\left(2Q + D_1\left(-QT^2e_h + 2de_T\right)\right)\right)\right) \end{bmatrix} \tag{B.1.7}$$

Since $\frac{dc_2e_T}{Q^4T^2}$ is positive, then the equation (B.1.7) is positive if,

$$\begin{bmatrix} \left(QT(-2p + Tc_h + 2c_p)D_1 + c_2\left(2Q + D_1\left(-QT^2e_h + 2de_T\right)\right)\right) \\ \left(-d(a - 2bG)^2c_2D_1^2e_T - 2b(mv + D_0 + G(a - bG)D_1)\right) \end{bmatrix} > 0 \tag{B.1.8}$$

Finally, if the conditions (B.1.6) and (B.1.8) are satisfied, the total profit is always concave.

Proposition 2 discusses the usefulness of investing in carbon emission reduction.

Appendix B.2

Proof of Proposition 2:

For,

$TP_2(Q,\ G) - TP_2(Q,\ 0)$

$$= \begin{bmatrix} \frac{1}{T}\left(pDT - C_h\frac{DT^2}{2} - C_pDT - \right. \\ C_2\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - \ R(G)\right) + C_2U - \ G - OC\Big) \\ -\frac{1}{T}\left(pDT - C_h\frac{DT^2}{2} - C_pDT - \ C_2\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2}\right) + C_2U - OC\right) \end{bmatrix}$$

$$= \frac{1}{T}[c_2R(G) - G]$$

$$= \frac{1}{T}\left[c_2\left(aG - bG^2\right) - G\right] \tag{B.2.1}$$

At the optimal point $(Q^*,\ G^*)$,

$TP_2(Q^*,\ G^*) - TP_2(Q^*,\ 0)$

$$= \frac{1}{T}\begin{bmatrix} c_2\left(\begin{array}{c} -\frac{\left\{adc_2D_1E_T \pm \sqrt{a^2d^2c_2^2D_1^2{E_T}^2 + 4bdc_2D_1E_T(dmvc_2E_T + dc_2D_0E_T)}\right\}^2}{4bd^2c_2^2D_1^2{E_T}^2} \\ + \frac{a\left\{adc_2D_1E_T \pm \sqrt{a^2d^2c_2^2D_1^2{E_T}^2 + 4bdc_2D_1E_T(dmvc_2E_T + dc_2D_0E_T)}\right\}}{2bdc_2D_1E_T} \end{array}\right) \\ -\frac{adc_2D_1E_T \pm \sqrt{a^2d^2c_2^2D_1^2{E_T}^2 + 4bdc_2D_1E_T(dmvc_2E_T + dc_2D_0E_T)}}{2bdc_2D_1E_T} \end{bmatrix}$$

Simplifying, one has

$TP_2(Q^*,\ G^*) - 2(Q^*,\ 0)$

$$= \frac{1}{T}\begin{bmatrix} \left(\left(adc_2D_1E_T \pm \sqrt{d^2c_2^2D_1\left(4bmv + 4bD_0 + a^2D_1\right)E_T^2}\right) + \right. \\ 2d(1 - ac_2)D_1E_T\Big)\left(\frac{1}{4bd^2c_2D_1^2E_T^2}\right. \\ \left.\pm\sqrt{d^2c_2^2D_1E_T^2\left(4bmv + 4bD_0 + a^2D_1\right)} - adc_2D_1E_T\right) \end{bmatrix} \tag{B.2.2}$$

From the value $\beta_3 = adc_2E_TD_1$; $\beta_4 = bdc_2E_TD_1$ and $\beta_5 = dc_2E_TD_0 + dc_2E_Tmv$ in the equation (B.2.2), one has:

$TP_2(Q^*,\ G^*) - TP_2(Q^*,\ 0)$

$$= \frac{1}{T}\begin{bmatrix} c_2\left(\begin{array}{c} -\frac{b\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)^2}{4\beta_4^2} + \frac{a\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_4} \\ -\frac{\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}}{2\beta_4} \end{array}\right) \end{bmatrix} > 0$$

Thus, one has

$$E_2(Q,\ 0) - E_2(Q,\ G) = \begin{bmatrix} \frac{1}{T}\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2}\right) \\ -\frac{1}{T}\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right) \end{bmatrix}$$

$$= \frac{1}{T}[R(G)]$$

$$= \frac{1}{T}\left[aG - bG^2\right] \tag{B.2.3}$$

At the optimal point $(Q^*,\ G^*)$,

$E_2(Q^*,\ 0) - E_2(Q^*,\ G^*)$

$$= \frac{1}{T}\begin{bmatrix} -\frac{1}{4bd^2c_2^2D_1^2{E_T}^2}(adc_2D_1E_T \pm \\ \sqrt{\left(a^2d^2c_2^2D_1^2{E_T}^2 + 4bdc_2D_1E_T(dmvc_2E_T + dc_2D_0E_T)\right)}\Big)^2 + \frac{a}{2bdc_2D_1E_T} \\ \left(adc_2D_1E_T \pm \sqrt{a^2d^2c_2^2D_1^2{E_T}^2 + 4bdc_2D_1E_T(dmvc_2E_T + dc_2D_0E_T)}\right) \end{bmatrix}$$

Simplifying, one has

$E_2(Q^*,\ 0) - E_2(Q^*,\ G^*) =$

$$\frac{1}{T}\begin{pmatrix} \frac{1}{4bd^2c_2^2D_1^2E_T^2} \\ \left(\pm\sqrt{d^2c_2^2D_1\left(4bmv + 4bD_0 + a^2D_1\right)E_T^2} - adc_2D_1E_T\right) \\ \left(\left(adc_2D_1E_T \pm \sqrt{d^2c_2^2D_1\left(4bmv + 4bD_0 + a^2D_1\right)E_T^2}\right) \right. \\ \left. -2adc_2D_1E_T\right) \end{pmatrix} \tag{B.2.4}$$

From the values β_3, β_4 and β_5 in the equation (B.2.4), one has

$$E_2(Q^*,\ 0) - E_2(Q^*,\ G^*) = \frac{1}{T}\left[\frac{a\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)}{2\beta_4} - \frac{B\left(\beta_3 \pm \sqrt{\beta_3^2 + 4\beta_4\beta_5}\right)^2}{4\beta_4^2}\right] > 0$$

Therefore, investment in green technology is effective under the carbon trading policy.

Appendix C.1

Proof of Lemma 3:

The following procedure is used to prove the concavity of the function.

Step 1: Find $\frac{\partial TP_3(\psi,G)}{\partial G}$.

Differentiate Eq. (21) with respect to G

$$\frac{\partial TP_3(G,\psi)}{\partial G} = \frac{\partial}{\partial G}\left[\frac{1}{T}[-\phi_2 + 2G\phi_3 - \psi\phi_5 + 2G\psi\phi_6]\right] = 0 \tag{C.1.1}$$

Step 2: Find G^* by using **step 1**

Solve equation (C.1.1)

$$G^*(\psi) = \frac{\phi_2 + \psi\phi_5}{2(\phi_3 + 2\psi\phi_6)}$$

which is the optimal point.

Step 3: Find $\frac{\partial^2 TP_3(G,\psi)}{\partial G^2}$

Differentiate (C.1.1) with respect to G,

$$\frac{\partial^2 TP_3(G,\psi)}{\partial G^2} = 2\phi_3 + 2\psi\phi_6 \tag{C.1.2}$$

Step 4: Proof the optimality conditions for the objective function. from,

$$\frac{\partial^2 TP_3(G,\psi)}{\partial G^2} = 2\phi_3 + 2\psi\phi_6$$

Simplifying, one has

$$2b\psi + 2bTD_1 + bTD_1TC_h + 2bTD_1C_p - (TbTD_1\psi E_h + 2p)$$

which will be less than zero if

$$(TbTD_1\psi E_h + 2p) > 2b\psi + 2bTD_1 + bTD_1TC_h + 2bTD_1C_p \tag{C.1.3}$$

Finally, if the condition (C.1.3) is satisfied, the total profit is always concave.

Proposition 3 discusses the usefulness of investing in carbon emission reduction.

Appendix C.2

Proof of Proposition 3:

For,

$$TP_3(Q,\ G(\psi)) - TP_3(Q,\ 0) = \left[\begin{array}{l}\frac{1}{T}\left[pDT - C_h\frac{DT^2}{2} - C_pDT - G - OC - \psi\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G) - \right.\right.\\ \left.\left. W)\right] - \frac{1}{T}\left[pDT - C_h\frac{DT^2}{2} - C_pDT - OC - \psi\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - W\right)\right]\end{array}\right]$$

$$= \frac{1}{T}[\psi R(G) - G\ ]$$

$$= \frac{1}{T}\left[\psi\left(aG - bG^2\right) - G\right] \tag{C.2.1}$$

At the optimal point $(Q^*,\ G^*)$,

$$TP_3(Q^*,\ G^*) - TP_3(Q^*,\ 0) = \frac{1}{T}\left[\begin{array}{l}\psi\left(-\frac{b\left(1+\frac{1}{2}aT^2C_h^2D_1+\psi\left(a-\frac{1}{2}aT^2D_1E_h\right)-a(DpT_1)+a(D_1TC_p)\right)^2}{4\left(\frac{1}{2}bT^2C_h^2D_1-b(DpT_1)+b(D_1TC_p)+2\psi\left(b-\frac{1}{2}T^2(bE_hD_1)\right)\right)^2}\right.\\ \left.+\frac{a\left(1+\frac{1}{2}aT^2C_h^2D_1+\psi\left(a-\frac{1}{2}aT^2D_1E_h\right)-a(D_1pT)+a(D_1TC_p)\right)}{2\left(\frac{1}{2}bT^2C_h^2D_1-b(D_1pT)+b(D_1TC_p)+2\psi\left(b-\frac{1}{2}T^2(bE_hD_1)\right)\right)}\right)\\ -\frac{1+\frac{1}{2}aT^2C_h^2D_1+\psi\left(a-\frac{1}{2}aT^2D_1E_h\right)-a(D_1pT)+a(D_1TC_p)}{2\left(\frac{1}{2}bT^2C_h^2D_1-b(D_1pT)+b(D_1TC_p)+2\psi\left(b-\frac{1}{2}T^2(bE_hD_1)\right)\right)}\end{array}\right]$$

Simplifying, one has

$$TP_3(Q^*,G^*) - TP_3(Q^*,0) = \frac{1}{T}\left[\frac{\left(\begin{array}{l}\left(2 + 2a\psi + aT^2C_h^2D_1 - aT^2\psi D_1E_h - 2a(DpT_1) + \right.\\ 2a(D_1TC_p))\left(-10b\psi + 6ab\psi^2 + bT^2(-2 + a\psi)C_h^2D_1 + \right.\\ abT^2\psi^2D_1E_h + 4b(D_1pT) - 2ab\psi(D_1pT) - 4b(D_1TC_p)\\ \left.\left.+2ab\psi(D_1TC_p) + 4T^2\psi(bD_1E_h) - 4aT^2\psi^2(bE_hD_1)\right)\right)\end{array}}{\left(4\left(4b\psi + bT^2C_h^2D_1 - 2b(D_1pT) + 2b(D_1TC_p) - 2T^2\psi(bE_hD_1)\right)^2\right)}\right] \tag{C.2.2}$$

From

$$\phi_2 = \frac{D_1aC_hT^2}{2} + C_pTD_1a + 1 - pTD_1a;$$

$$\phi_3 = \frac{D_1bC_hT^2}{2} - pTD_1b + C_pTD_1b;$$

$$\phi_5 = a - \frac{E_hT^2aD_1}{2} \text{ and } \phi_6 = b - \frac{D_1bE_hT^2}{2}$$

in the equation (C.2.2), one has

$$TP_3(Q^*,\ G^*) - TP_3(Q^*,\ 0) = \frac{1}{T}\left[\psi\left(-\frac{b(\phi_2+\psi\phi_5)^2}{4(\phi_3+2\psi\phi_6)^2} + \frac{a(\phi_2+\psi\phi_5)}{2(\phi_3+2\psi\phi_6)}\right) - \frac{\phi_2+\psi\phi_5}{2(\phi_3+2\psi\phi_6)}\right] > 0$$

One has,

$$E_3(Q,\ 0) - E_3(Q,\ G) = \left[\begin{array}{c}\frac{1}{T}\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2}\right)\\ -\frac{1}{T}\left(\frac{D}{Q}E_Td + E_h\frac{DT^2}{2} - R(G)\right)\end{array}\right]$$

$$= \frac{1}{T}[R(G)]$$

$$= \frac{1}{T}\left[aG - bG^2\right] \tag{C.2.3}$$

At the optimal point $(Q^*,\ G^*)$,

$$E_3(Q^*,\ 0) - E_3(Q^*,\ G^*) = \frac{1}{T}\left[\frac{\left(\begin{array}{l}\left(4\left(4b\psi + bT^2C_h^2D_1 - 2b(D_1pT) + 2b(D_1TC_p) - \right.\right.\\ \left.2T^2\psi(bE_hD_1)\right)^2(6ab\psi + abT^2C_h^2D_1 + abT^2\psi D_1E_h - \\ 2b - 2ab(D_1pT) + 2ab(D_1TC_p) - 4aT^2\psi(bE_hD_1)\right)\end{array}}{4\left(4b\psi + bT^2C_h^2D_1 - 2b(D_1pT) + 2b(D_1TC_p) - 2T^2\psi(bE_hD_1)\right)^2}\right] \tag{C.2.4}$$

Substituting the value of ϕ_2, ϕ_3, ϕ_5 and ϕ_6 in equation (C.2.4) and simplifying, one has:

$$E_3(Q^*,\ 0) - E_3(Q^*,\ G^*) = \frac{1}{T}\left[\frac{a(\phi_2+\psi\phi_5)}{2(\phi_3+2\psi\phi_6)} - \frac{b(\phi_2+\psi\phi_5)^2}{4(\phi_3+2\psi\phi_6)^2}\right] > 0$$

Therefore, investment in green technology is effective under limited carbon.

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